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PERSONALIZED AVATAR NOTIFICATION

Abstract

An avatar notification system is disclosed, which performs operations that include: causing display of a notification at a client device associated with a first user account, the notification including an identification of a second user account; receiving an input that selects the notification from the client device; presenting a composition interface at the client device in response to the input that selects the notification, the composition interface including a display of a media element that comprises a first identifier associated with the first user account and a second identifier associated with the second user account; receiving a selection of the media element from the client device; and generating a message that includes the media element in response to the selection.

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Background/Summary

CLAIM OF PRIORITY [0001] This application is a continuation of U.S. patent application Ser. No. 18/153,750, filed Jan. 12, 2023, which application is a continuation of U.S. patent application Ser. No. 17/343,569, filed Jun. 9, 2021, now issued as U.S. Pat. No. 11,563,702, which application is a continuation of U.S. patent application Ser. No. 16/702,184, filed Dec. 3, 2019, now issued as U.S. Pat. No. 11,063,891, which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] Embodiments of the present disclosure relate generally to mobile computing technology and, more particularly, but not by way of limitation, to systems for generating and presenting notifications at client devices.

BACKGROUND

[0003] An avatar is a graphical representation of a user, or a user's digital presence. It may take either a three-dimensional form, as in games or virtual worlds, or a two-dimensional form as an icon as typically seen in internet forums and online communities. Users are often allowed to select an avatar from a preset list of avatars or may be presented with interfaces to personalize an avatar based on selections of avatar attributes.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0004] To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

[0005] FIG. 1 is a block diagram showing an example messaging system for exchanging data (e.g., messages and associated content) over a network in accordance with some embodiments, wherein the messaging system includes an avatar notification system 124.

[0006] FIG. 2 is block diagram illustrating further details regarding a messaging system, according to example embodiments.

[0007] FIG. 3 is a block diagram illustrating various modules of an avatar notification system 124, according to certain example embodiments.

[0008] FIG. 4 is a flowchart depicting a method of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments.

[0009] FIG. 5 is a flowchart depicting a method of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments.

[0010] FIG. 6 is a flowchart depicting a method of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments.

[0011] FIG. 7 is a flowchart depicting a method of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments.

[0012] FIG. **8** is an interface diagram depicting graphical user interfaces (GUI) generated and displayed by an avatar notification system **124**, according to certain example embodiments.

[0013] FIG. **9** is a block diagram illustrating a representative software architecture, which may be used in conjunction with various hardware architectures herein described and used to implement various embodiments.

[0014] FIG. **10** is a block diagram illustrating components of a machine, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

DETAILED DESCRIPTION

[0015] As discussed above, users of various social networks and online communities may select avatars to be used as graphical representations of themselves within the online communities. Accordingly, the selection of a user's avatar is often considered a significant, or notable occasion within an online community. Accordingly, in certain example embodiments, an avatar notification system is disclosed, which performs operations that include: causing display of a notification at a client device associated with a first user account, the notification including an identification of a second user account; receiving an input that selects the notification from the client device; presenting a composition interface at the client device in response to the input that selects the notification, the composition interface including a display of a media element that comprises a first identifier associated with the first user account and a second identifier associated with the second user account; receiving a selection of the media element from the client device; and generating a message that includes the media element in response to the selection.

[0016] According to certain example embodiments, the avatar notification system **124** may present the notification at the client device associated with the first user account responsive to receiving a selection of a personalized avatar from the second user account. For example, the selection of the personalized avatar from the second user account may be the first time that a user associated with the second user account selects or generates a personalized avatar.

[0017] Responsive to receiving the selection of the personalized avatar from the second user account, the avatar notification system **124** access user profile data associated with the second user profile to identify one or more user accounts (i.e., the first user account). The one or more user accounts may be identified based on historical interaction data between the second user account and each of the one or more user accounts, wherein the historical interaction data includes bidirectional communications. For example, the avatar notification system **124** may identify users in which the user associated with the second user account has engaged in recent conversations with or may be close with based on the historical interaction data. The identification of the one or more users may therefore be based on a length of time in which the second user has been connected with (i.e., buddies) on a social networking platform, how frequently the user associated with the second user account engages in bidirectional communications with the one or more users, or how recently the second user communicated with the one or more users.

[0018] According to certain example embodiments, the composition interface presented by the avatar notification system **124** may include a display of a plurality of media elements, wherein each media element among the plurality of media elements comprises media attributes. In some embodiments, the media attributes of the plurality of media elements may be selected based on user profile data associated with the first user and the second user, such as the historical interaction data discussed above. The media elements may be generated as a composite of the personalized avatars of one or more users (i.e., the first user account and the second user account). As an illustrative example, the personalized avatar may include Bitmoji that resemble the users themselves or have an appearance configured based on user inputs. Each of the media elements may comprise a display of a Bitmoji associated with the first user account and the second user account, in various poses or presentations. For example, a media element may depict Bitmojis of the first and second user account engaged in a conversation, or some other activity, wherein the activity may be depicted

based on a set of display attributes, selected based on the historical interaction data.

[0019] For example, if the historical interaction data and user profile data indicates a familial relationship between the first user account and the second user account, the media elements may be selected to depict this familial relationship. Similarly, if the historical interaction data and the user profile data indicate that the first user account and the second user account both share certain interests (i.e., both users are members of certain clubs or social media groups, or follow user accounts and content providers related to certain hobbies or activities like cooking, painting, or running), then the media elements may be selected to depict those interests.

[0020] According to certain embodiments, a user may provide an input selecting one or more of the media elements from among the plurality of media elements presented in the composition interface. Responsive to receiving the input the avatar notification system **124** may generate a message that includes the selected media element, wherein the message includes an ephemeral message.

[0021] FIG. **1** is a block diagram showing an example messaging system **100** for exchanging data (e.g., messages and associated content) over a network. The messaging system **100** includes one or more client device **102** which host a number of applications including a messaging client application **104**. Each messaging client application **104** is communicatively coupled to other instances of the messaging client application **104** and a messaging server system **108** via a network **106** (e.g., the Internet).

[0022] Accordingly, each messaging client application **104** is able to communicate and exchange data with another messaging client application **104** and with the messaging server system **108** via the network **106**. The data exchanged between messaging client applications **104**, and between a messaging client application **104** and the messaging server system **108**, includes functions (e.g., commands to invoke functions) as well as payload data (e.g., text, audio, video or other multimedia data).

[0023] The messaging server system **108** provides server-side functionality via the network **106** to a particular messaging client application **104**. While certain functions of the messaging system **100** are described herein as being performed by either a messaging client application **104** or by the messaging server system **108**, it will be appreciated that the location of certain functionality either within the messaging client application **104** or the messaging server system **108** is a design choice. For example, it may be technically preferable to initially deploy certain technology and functionality within the messaging server system **108**, but to later migrate this technology and functionality to the messaging client application **104** where a client device **102** has a sufficient processing capacity.

[0024] The messaging server system **108** supports various services and operations that are provided to the messaging client application **104**. Such operations include transmitting data to, receiving data from, and processing data generated by the messaging client application **104**. In some embodiments, this data includes, message content, client device information, geolocation information, media annotation and overlays, message content persistence conditions, social network information, and live event information, as examples. In other embodiments, other data is used. Data exchanges within the messaging system **100** are invoked and controlled through functions available via GUIs of the messaging client application **104**.

[0025] Turning now specifically to the messaging server system **108**, an Application Program Interface (API) server **110** is coupled to, and provides a programmatic interface to, an application server **112**. The application server **112** is communicatively coupled to a database server **118**, which facilitates access to a database **120** in which is stored data associated with messages processed by the application server **112**.

[0026] Dealing specifically with the Application Program Interface (API) server **110**, this server receives and transmits message data (e.g., commands and message payloads) between the client device **102** and the application server **112**. Specifically, the Application Program Interface (API) server **110** provides a set of interfaces (e.g., routines and protocols) that can be called or queried by

the messaging client application **104** in order to invoke functionality of the application server **112**. The Application Program Interface (API) server **110** exposes various functions supported by the application server **112**, including account registration, login functionality, the sending of messages, via the application server **112**, from a particular messaging client application **104** to another messaging client application **104**, the sending of media files (e.g., images or video) from a messaging client application **104** to the messaging server application **114**, and for possible access by another messaging client application **104**, the setting of a collection of media data (e.g., story), the retrieval of a list of friends of a user of a client device **102**, the retrieval of such collections, the retrieval of messages and content, the adding and deletion of friends to a social graph, the location of friends within a social graph, opening and application event (e.g., relating to the messaging client application **104**).

[0027] The application server **112** hosts a number of applications and subsystems, including a messaging server application **114**, an image processing system **116**, a social network system **122**, and an avatar notification system **124**. The avatar notification system **124** is configured to generate and present a notification in response to detecting a personalization of a user avatar by a user account. Further details of the avatar notification system **124** can be found in FIG. 3 below.

[0028] The messaging server application **114** implements a number of message processing technologies and functions, particularly related to the aggregation and other processing of content (e.g., textual and multimedia content) included in messages received from multiple instances of the messaging client application **104**. As will be described in further detail, the text and media content from multiple sources may be aggregated into collections of content (e.g., called stories or galleries). These collections are then made available, by the messaging server application **114**, to the messaging client application **104**. Other processor and memory intensive processing of data may also be performed server-side by the messaging server application **114**, in view of the hardware requirements for such processing.

[0029] The application server **112** also includes an image processing system **116** that is dedicated to performing various image processing operations, typically with respect to images or video received within the payload of a message at the messaging server application **114**.

[0030] The social network system **122** supports various social networking functions services, and makes these functions and services available to the messaging server application **114**. To this end, the social network system **122** maintains and accesses an entity graph **304** within the database **120**. Examples of functions and services supported by the social network system **122** include the identification of other users of the messaging system **100** with which a particular user has relationships or is “following,” and also the identification of other entities and interests of a particular user.

[0031] The application server **112** is communicatively coupled to a database server **118**, which facilitates access to a database **120** in which is stored data associated with messages processed by the messaging server application **114**.

[0032] FIG. 2 is block diagram illustrating further details regarding the messaging system **100**, according to example embodiments. Specifically, the messaging system **100** is shown to comprise the messaging client application **104** and the application server **112**, which in turn embody a number of some subsystems, namely an ephemeral timer system **202**, a collection management system **204** and an annotation system **206**.

[0033] The ephemeral timer system **202** is responsible for enforcing the temporary access to content permitted by the messaging client application **104** and the messaging server application **114**. To this end, the ephemeral timer system **202** incorporates a number of timers that, based on duration and display parameters associated with a message, collection of messages (e.g., a SNAPCHAT story), or graphical element, selectively display and enable access to messages and associated content via the messaging client application **104**. Further details regarding the operation of the ephemeral timer system **202** are provided below.

[0034] The collection management system **204** is responsible for managing collections of media (e.g., collections of text, image video and audio data). In some examples, a collection of content (e.g., messages, including images, video, text and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **204** may also be responsible for publishing an icon that provides notification of the existence of a particular collection to the user interface of the messaging client application **104**.

[0035] The collection management system **204** furthermore includes a curation interface **208** that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface **208** enables an event organizer to curate a collection of content relating to a specific event (e.g., delete inappropriate content or redundant messages). Additionally, the collection management system **204** employs machine vision (or image recognition technology) and content rules to automatically curate a content collection. In certain embodiments, compensation may be paid to a user for inclusion of user generated content into a collection. In such cases, the curation interface **208** operates to automatically make payments to such users for the use of their content.

[0036] The annotation system **206** provides various functions that enable a user to annotate or otherwise modify or edit media content associated with a message. For example, the annotation system **206** provides functions related to the generation and publishing of media overlays for messages processed by the messaging system **100**. The annotation system **206** operatively supplies a media overlay (e.g., a SNAPCHAT filter, lens) to the messaging client application **104** based on a geolocation of the client device **102**. In another example, the annotation system **206** operatively supplies a media overlay to the messaging client application **104** based on other information, such as, social network information of the user of the client device **102**. A media overlay may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects, as well as animated facial models. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo or video) at the client device **102**. For example, the media overlay including text that can be overlaid on top of a photograph generated taken by the client device **102**. In another example, the media overlay includes an identification of a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In another example, the annotation system **206** uses the geolocation of the client device **102** to identify a media overlay that includes the name of a merchant at the geolocation of the client device **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the database **120** and accessed through the database server **118**.

[0037] In one example embodiment, the annotation system **206** provides a user-based publication platform that enables users to select a geolocation on a map, and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The annotation system **206** generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

[0038] In another example embodiment, the annotation system **206** provides a merchant-based publication platform that enables merchants to select a particular media overlay associated with a geolocation via a bidding process. For example, the annotation system **206** associates the media overlay of a highest bidding merchant with a corresponding geolocation for a predefined amount of time

[0039] FIG. **3** is a block diagram illustrating components of the avatar notification system **124** that configure the avatar notification system **124** to perform operations to generate and cause display of

a notification at one or more client devices (i.e., client device **102**) responsive to detecting the personalization or selection of an avatar by a user account, according to certain example embodiments.

[0040] In further embodiments, the components of the avatar notification system **124** may configure the avatar notification system **124** to perform operations that include: causing display of a notification at a client device associated with a first user account, the notification including an identification of a second user account; receiving an input that selects the notification from the client device; presenting a composition interface at the client device in response to the input that selects the notification, the composition interface including a display of a media element that comprises a first identifier associated with the first user account and a second identifier associated with the second user account; receiving a selection of the media element from the client device; and generating a message that includes the media element in response to the selection.

[0041] The avatar notification system **124** is shown as including a user profile module **302**, a notification module **304**, and a presentation module **306**, all configured to communicate with each other (e.g., via a bus, shared memory, or a switch). Any one or more of these modules may be implemented using one or more processors **308** (e.g., by configuring such one or more processors to perform functions described for that module) and hence may include one or more of the processors **308**. In certain embodiments, the avatar notification system **124** may include or have access to the database **120**, wherein the database **120** may comprise a collection of media content indexed based on user attributes and astrological signs.

[0042] Any one or more of the modules described may be implemented using hardware alone (e.g., one or more of the processors **308** of a machine) or a combination of hardware and software. For example, any module described of the avatar notification system **124** may physically include an arrangement of one or more of the processors **308** (e.g., a subset of or among the one or more processors of the machine) configured to perform the operations described herein for that module. As another example, any module of the avatar notification system **124** may include software, hardware, or both, that configure an arrangement of one or more processors **308** (e.g., among the one or more processors of the machine) to perform the operations described herein for that module. Accordingly, different modules of the avatar notification system **124** may include and configure different arrangements of such processors **308** or a single arrangement of such processors **308** at different points in time. Moreover, any two or more modules of the avatar notification system **124** may be combined into a single module, and the functions described herein for a single module may be subdivided among multiple modules. Furthermore, according to various example embodiments, modules described herein as being implemented within a single machine, database, or device may be distributed across multiple machines, databases, or devices.

[0043] FIG. **4** is a flowchart depicting a method **400** of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments. Operations of the method **400** may be performed by the modules described above with respect to FIG. **3**. As shown in FIG. **4**, the method **400** includes one or more operations **402**, **404**, **406**, **408**, and **410**.

[0044] At operation **402**, the notification module **304** causes display of a notification that includes an identification of a second user account at a client device **102**, wherein the client device **102** is associated with a first user account. The identification of the second user account may include an identifier associated with the second user account (i.e., a username), as well as a personalized avatar selected by a user associated with the second user account. In some example embodiments, as described in the method **500** of FIG. **5**, the notification module **304** may present the notification (e.g., the notification **815** of FIG. **8**) responsive to detecting a selection or personalization of an avatar to be associated with a user account (i.e., the second user account).

[0045] At operation **404**, the presentation module **306** receives an input that selects the notification from the client device **102**. For example, a user associated with a first user account may provide a tactile input that selects the notification at the client device **102**. Responsive to receiving the input

that selects the notification, at operation **406**, the presentation module **306** causes display of a composition interface at the client device **102**, wherein the composition interface includes a display of one or more media elements. According to certain example embodiments, the media elements may comprise a composite image generated based on a first user identifier associated with the first user account, and a second user identifier associated with the second user account, wherein the user identifiers may include personalized avatars, such as Bitmoji.

[0046] Accordingly, the media elements may comprise media properties, wherein the media properties may be selected based on a relationship between the first user account and the second user account, wherein the relationship may be determined based on an interaction history that comprises at least bidirectional communications between the first user account and the second user account. For example, based on an analysis of user profile data, as well as the bidirectional communications, the avatar notification system may determine a relationship status or type associated with the first user account and the second user account. Factors that may be taken into consideration include explicit indicators of relationship status, or based on a frequency of bidirectional communication, a recency of bidirectional communication, as well as user profile data indicating shared user groups or indications of common user accounts followed by both users.

[0047] At operation **408**, the presentation module **306** receives a selection of a media element from among the one or more media elements presented within the composition interface at the client device **102**. Responsive to receiving the selection of the media element, at operation **410** the notification module **308** generates a message that includes the selected media element, wherein the message is addressed to the second user account.

[0048] FIG. **5** is a flowchart depicting a method **500** of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments. Operations of the method **500** may be performed by the modules described above with respect to FIG. **3**. In some embodiments, the method **500** may be performed as part of (i.e., a subroutine) operation **402** of the method **400**. As shown in FIG. **5**, the method **500** includes one or more operations **502**, **504**, and **506**.

[0049] At operation **502**, the user profile module **302** receives a selection or personalization of an avatar (i.e., the second user identifier) from the second user account. For example, a user of the second user account may provide inputs that configure a personalized avatar, by selecting various features and attributes of the personalized avatar.

[0050] At operation **504**, responsive to receiving the selection or personalization of the avatar, the user profile module **302** identifies one or more user accounts associated with the second user account. For example, the user profile module **302** may identify one or more user accounts that have participated in a bidirectional communication with the second user account within a threshold period of time or based on a relationship status associated with the second user account and the one or more user accounts.

[0051] Responsive to identifying the one or more user accounts, the notification module **304** causes display of the notification at the one or more user accounts (that include the first user account), as discussed in operation **402** of the method **400**, and as can be seen in the interface **810** depicted in FIG. **8**.

[0052] FIG. **6** is a flowchart depicting a method **600** of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments. Operations of the method **600** may be performed by the modules described above with respect to FIG. **3**. In some embodiments, the method **600** may be performed as part of (i.e., a subroutine) operation **406** of the method **400**. As shown in FIG. **6**, the method **600** includes one or more operations **602** and **604**.

[0053] At operation **602**, responsive to identifying one or more user accounts that include a first user account, as in operation **504** of the method **500**, the user profile module **302** accesses interaction history associated with the second user account and the one or more user accounts, wherein the interaction history comprises bidirectional communications and interactions between

the second user account and the one or more user accounts. For example, the bidirectional communications may comprise messages sent between the user accounts, wherein the messages include timestamps and message content that may include media items. In such embodiments, the media items may themselves comprise media properties that may be used to determine a relationship status between the second user account and the one or more user accounts. For example, frequent use of a “heart” emoji in communications may provide an indication of a romantic relationship.

[0054] At operation **604**, the presentation module **306** selects media properties of the one or more media elements presented within the composition interfaces at each of the client devices associated with one or more user accounts based on the user profile data and interaction history associated with the second user and each of the one or more user accounts. As an illustrative example, the avatar notification system **124** may characterize a relationship of the second user account and each of the one or more user accounts based on the corresponding interaction histories. As discussed above, a characterization may be “romantic,” “familial,” “co-worker,” or simply just “close friend.” Based on the characterization, the presentation module **306** may select media properties of the media elements to be presented in the corresponding composition interfaces at the one or more client devices associated with each of the one or more user accounts.

[0055] FIG. **7** is a flowchart depicting a method **700** of presenting a notification based on a selection of a personalized avatar, according to certain example embodiments. Operations of the method **700** may be performed by the modules described above with respect to FIG. **3**. In certain embodiments, the method **700** may be performed as a subroutine of operation **402** of the method **400** depicted in FIG. **4**. As shown in FIG. **7**, the method **700** includes one or more operations **702**, **704**, and **706**.

[0056] At operation **702**, the user profile module **302** accesses an interaction history associated with the first user account and a plurality of user accounts from a list of user connections associated with the second user account, wherein the plurality of user accounts include the second user account, wherein the interaction history comprises at least bidirectional communications between the first user account and the second user account, and wherein the bidirectional communications include timestamps.

[0057] At operation **704**, the user profile module **302** determines that a timestamp from among a plurality of timestamps associated with a bidirectional communication between the second user account and the first user account transgresses a threshold value. In such embodiments, the transgression of the threshold value may indicate that a bidirectional communication from among the bidirectional communications occurred within a threshold period of time (i.e., within a week, a month, etc.).

[0058] Responsive to determining that the timestamp associated with the bidirectional communication between the second user account and the first user account transgresses the threshold value, at operation **706** (as in operation **402**), the notification module **304** causes display of the notification that includes the identification of the second user account at the client device **102** associated with the first user account.

[0059] FIG. **8** is an interface diagram **800** depicting GUIs generated and displayed by an avatar notification system **124**, according to certain example embodiments. As seen in FIG. **8**, the interface diagram **800** comprises a GUI **805** and a GUI **810**.

[0060] GUI **805** includes a display of a notification **815**, as discussed in operation **402** of the method **400**. As seen in FIG. **8**, the notification **815** may include an identification of a second user account (i.e., “Laurent”).

[0061] GUI **810** includes a display of a composition interface, wherein the composition interface comprises a presentation of one or more media elements **820**, wherein attributes of the media elements **820** may be based on personalized avatars associated with the first user account and the second user account, and a characterization of a relationship between the first user account and the

second user account.

[0062] Accordingly, a user of the client device **102** may provide inputs that select one or more of the media elements from among the media elements **820**, and in response, the avatar notification system **124** may generate a message addressed to the second user account that includes the selected media elements, wherein the message includes an ephemeral message.

Software Architecture

[0063] FIG. **9** is a block diagram illustrating an example software architecture **906**, which may be used in conjunction with various hardware architectures herein described. FIG. **9** is a non-limiting example of a software architecture and it will be appreciated that many other architectures may be implemented to facilitate the functionality described herein. The software architecture **906** may execute on hardware such as machine **1000** of FIG. **10** that includes, among other things, processors **1004**, memory **1014**, and I/O components **1018**. A representative hardware layer **952** is illustrated and can represent, for example, the machine **900** of FIG. **9**. The representative hardware layer **952** includes a processing unit **954** having associated executable instructions **904**. Executable instructions **904** represent the executable instructions of the software architecture **906**, including implementation of the methods, components and so forth described herein. The hardware layer **952** also includes memory and/or storage modules memory/storage **956**, which also have executable instructions **904**. The hardware layer **952** may also comprise other hardware **958**.

[0064] In the example architecture of FIG. **9**, the software architecture **906** may be conceptualized as a stack of layers where each layer provides particular functionality. For example, the software architecture **906** may include layers such as an operating system **902**, libraries **920**, applications **916** and a presentation layer **914**. Operationally, the applications **916** and/or other components within the layers may invoke application programming interface (API) API calls **908** through the software stack and receive a response as in response to the API calls **908**. The layers illustrated are representative in nature and not all software architectures have all layers. For example, some mobile or special purpose operating systems may not provide a frameworks/middleware **918**, while others may provide such a layer. Other software architectures may include additional or different layers.

[0065] The operating system **902** may manage hardware resources and provide common services. The operating system **902** may include, for example, a kernel **922**, services **924** and drivers **926**. The kernel **922** may act as an abstraction layer between the hardware and the other software layers. For example, the kernel **922** may be responsible for memory management, processor management (e.g., scheduling), component management, networking, security settings, and so on. The services **924** may provide other common services for the other software layers. The drivers **926** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **926** include display drivers, camera drivers, Bluetooth® drivers, flash memory drivers, serial communication drivers (e.g., Universal Serial Bus (USB) drivers), Wi-Fi® drivers, audio drivers, power management drivers, and so forth depending on the hardware configuration.

[0066] The libraries **920** provide a common infrastructure that is used by the applications **916** and/or other components and/or layers. The libraries **920** provide functionality that allows other software components to perform tasks in an easier fashion than to interface directly with the underlying operating system **902** functionality (e.g., kernel **922**, services **924** and/or drivers **926**). The libraries **920** may include system libraries **944** (e.g., C standard library) that may provide functions such as memory allocation functions, string manipulation functions, mathematical functions, and the like. In addition, the libraries **920** may include API libraries **946** such as media libraries (e.g., libraries to support presentation and manipulation of various media format such as MPREG4, H.264, MP3, AAC, AMR, JPG, PNG), graphics libraries (e.g., an OpenGL framework that may be used to render 2D and 3D in a graphic content on a display), database libraries (e.g., SQLite that may provide various relational database functions), web libraries (e.g., WebKit that may provide web browsing functionality), and the like. The libraries **920** may also include a wide

variety of other libraries **948** to provide many other APIs to the applications **916** and other software components/modules.

[0067] The frameworks/middleware **918** (also sometimes referred to as middleware) provide a higher-level common infrastructure that may be used by the applications **916** and/or other software components/modules. For example, the frameworks/middleware **918** may provide various graphic user interface (GUI) functions, high-level resource management, high-level location services, and so forth. The frameworks/middleware **918** may provide a broad spectrum of other APIs that may be utilized by the applications **916** and/or other software components/modules, some of which may be specific to a particular operating system **902** or platform.

[0068] The applications **916** include built-in applications **938** and/or third-party applications **940**. Examples of representative built-in applications **938** may include, but are not limited to, a contacts application, a browser application, a book reader application, a location application, a media application, a messaging application, and/or a game application. Third-party applications **940** may include an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform, and may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or other mobile operating systems. The third-party applications **940** may invoke the API calls **908** provided by the mobile operating system (such as operating system **902**) to facilitate functionality described herein.

[0069] The applications **916** may use built in operating system functions (e.g., kernel **922**, services **924** and/or drivers **926**), libraries **920**, and frameworks/middleware **918** to create user interfaces to interact with users of the system. Alternatively, or additionally, in some systems interactions with a user may occur through a presentation layer, such as presentation layer **914**. In these systems, the application/component “logic” can be separated from the aspects of the application/component that interact with a user.

[0070] FIG. **10** is a block diagram illustrating components of a machine **1000**, according to some example embodiments, able to read instructions from a machine-readable medium (e.g., a machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **10** shows a diagrammatic representation of the machine **1000** in the example form of a computer system, within which instructions **1010** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1000** to perform any one or more of the methodologies discussed herein may be executed. As such, the instructions **1010** may be used to implement modules or components described herein. The instructions **1010** transform the general, non-programmed machine **1000** into a particular machine **1000** programmed to carry out the described and illustrated functions in the manner described. In alternative embodiments, the machine **1000** operates as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1000** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smart phone, a mobile device, a wearable device (e.g., a smart watch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1010**, sequentially or otherwise, that specify actions to be taken by machine **1000**. Further, while only a single machine **1000** is illustrated, the term “machine” shall also be taken to include a collection of machines that individually or jointly execute the instructions **1010** to perform any one or more of the methodologies discussed herein.

[0071] The machine **1000** may include processors **1004**, memory memory/storage **1006**, and I/O components **1018**, which may be configured to communicate with each other such as via a bus

1002. The memory/storage **1006** may include memory **1014**, such as a main memory, or other memory storage, and a storage unit **1016**, both accessible to the processors **1004** such as via the bus **1002**. The storage unit **1016** and memory **1014** store the instructions **1010** embodying any one or more of the methodologies or functions described herein. The instructions **1010** may also reside, completely or partially, within the memory **1014**, within the storage unit **1016**, within at least one of the processors **1004** (e.g., within the processor's cache memory), or any suitable combination thereof, during execution thereof by the machine **1000**. Accordingly, the memory **1014**, the storage unit **1016**, and the memory of processors **1004** are examples of machine-readable media.

[0072] The I/O components **1018** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1018** that are included in a particular machine **1000** will depend on the type of machine. For example, portable machines such as mobile phones will likely include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1018** may include many other components that are not shown in FIG. **10**. The I/O components **1018** are grouped according to functionality merely for simplifying the following discussion and the grouping is in no way limiting. In various example embodiments, the I/O components **1018** may include output components **1026** and input components **1028**. The output components **1026** may include visual components (e.g., a display such as a plasma display panel (PDP), a light emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The input components **1028** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and/or force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

[0073] In further example embodiments, the I/O components **1018** may include biometric components **1030**, motion components **1034**, environmental environment components **1036**, or position components **1038** among a wide array of other components. For example, the biometric components **1030** may include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components **1034** may include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The environment components **1036** may include, for example, illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometer that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment. The position components **1038** may include location sensor components (e.g., a Global Position system (GPS) receiver component), altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

[0074] Communication may be implemented using a wide variety of technologies. The I/O

components **1018** may include communication components **1040** operable to couple the machine **1000** to a network **1032** or devices **1020** via coupling **1022** and coupling **1024** respectively. For example, the communication components **1040** may include a network interface component or other suitable device to interface with the network **1032**. In further examples, communication components **1040** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1020** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a Universal Serial Bus (USB)).

[0075] Moreover, the communication components **1040** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1040** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1040**, such as, location via Internet Protocol (IP) geo-location, location via Wi-Fi® signal triangulation, location via detecting a NFC beacon signal that may indicate a particular location, and so forth.

GLOSSARY

[0076] “CARRIER SIGNAL” in this context refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible medium to facilitate communication of such instructions. Instructions may be transmitted or received over the network using a transmission medium via a network interface device and using any one of a number of well-known transfer protocols.

[0077] “CLIENT DEVICE” in this context refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smart phones, tablets, ultra books, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

[0078] “COMMUNICATIONS NETWORK” in this context refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other type of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others

defined by various standard setting organizations, other long range protocols, or other data transfer technology.

[0079] “EMPHEMERAL MESSAGE” in this context refers to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

[0080] “MACHINE-READABLE MEDIUM” in this context refers to a component, device or other tangible media able to store instructions and data temporarily or permanently and may include, but is not be limited to, random-access memory (RAM), read-only memory (ROM), buffer memory, flash memory, optical media, magnetic media, cache memory, other types of storage (e.g., Erasable Programmable Read-Only Memory (EEPROM)) and/or any suitable combination thereof. The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, or associated caches and servers) able to store instructions. The term “machine-readable medium” shall also be taken to include any medium, or combination of multiple media, that is capable of storing instructions (e.g., code) for execution by a machine, such that the instructions, when executed by one or more processors of the machine, cause the machine to perform any one or more of the methodologies described herein. Accordingly, a “machine-readable medium” refers to a single storage apparatus or device, as well as “cloud-based” storage systems or storage networks that include multiple storage apparatus or devices. The term “machine-readable medium” excludes signals per se.

[0081] “COMPONENT” in this context refers to a device, physical entity or logic having boundaries defined by function or subroutine calls, branch points, application program interfaces (APIs), or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various example embodiments, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein. A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a Field-Programmable Gate Array (FPGA) or an Application Specific Integrated Circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software) may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily

configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein. Considering embodiments in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In embodiments in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein. As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an Application Program Interface (API)). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other example embodiments, the processors or processor-implemented components may be distributed across a number of geographic locations.

[0082] “PROCESSOR” in this context refers to any circuit or virtual circuit (a physical circuit emulated by logic executing on an actual processor) that manipulates data values according to control signals (e.g., “commands”, “op codes”, “machine code”, etc.) and which produces corresponding output signals that are applied to operate a machine. A processor may, for example, be a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) processor, a Complex Instruction Set Computing (CISC) processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC) or any combination thereof. A processor may further be a

multi-core processor having two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously.

[0083] “TIMESTAMP” in this context refers to a sequence of characters or encoded information identifying when a certain event occurred, for example giving date and time of day, sometimes accurate to a small fraction of a second.

[0084] “LIFT” in this context is a measure of the performance of a targeted model at predicting or classifying cases as having an enhanced response (with respect to a population as a whole), measured against a random choice targeting model.

[0085] “PHONEME ALIGNMENT” in this context, a phoneme is a unit of speech that differentiates one word from another. One phoneme may consist of a sequence of closure, burst, and aspiration events; or, a diphthong may transition from a back vowel to a front vowel. A speech signal may therefore be described not only by what phonemes it contains, but also the locations of the phonemes. Phoneme alignment may therefore be described as a “time-alignment” of phonemes in a waveform, in order to determine an appropriate sequence and location of each phoneme in a speech signal.

[0086] “AUDIO-TO-VISUAL CONVERSION” in this context refers to the conversion of audible speech signals into visible speech, wherein the visible speech may include a mouth shape representative of the audible speech signal.

[0087] “TIME DELAYED NEURAL NETWORK (TDNN)” in this context, a TDNN is an artificial neural network architecture whose primary purpose is to work on sequential data. An example would be converting continuous audio into a stream of classified phoneme labels for speech recognition.

[0088] “BI-DIRECTIONAL LONG-SHORT TERM MEMORY (BLSTM)” in this context refers to a recurrent neural network (RNN) architecture that remembers values over arbitrary intervals. Stored values are not modified as learning proceeds. RNNs allow forward and backward connections between neurons. BLSTM are well-suited for the classification, processing, and prediction of time series, given time lags of unknown size and duration between events.

Claims

1. A method comprising: identifying a first user account and a second user account based on user profile data; determining a characterization of a relationship between the first user account and the second user account based on historical interaction data between the first user account and the second user account; causing display of a notification at a client device associated with the first user account; receiving an input that selects the notification from the client device; responsive to the input that selects the notification, generating a media item comprising media properties selected based on the characterization of the relationship between the first user account and the second user account; and presenting the media item at the client device.
2. The method of claim 1, wherein the historical interaction data comprises bidirectional communications between the first user account and the second user account.
3. The method of claim 1, wherein the notification includes a graphical avatar that identifies the second user account.
4. The method of claim 1, wherein the media item comprises a first graphical avatar representing the first user account and a second graphical avatar representing the second user account.
5. The method of claim 1, wherein presenting the media item includes: presenting a media collection that comprises a plurality of media items generated based on the first user account and the second user account, the plurality of media items including the media item.
6. The method of claim 1, wherein determining the characterization of the relationship includes: analyzing timestamps associated with the historical interaction data; and determining a frequency of interactions between the first user account and the second user account based on the timestamps.

7. The method of claim 1, wherein the media properties are selected based on shared interests between the first user account and the second user account identified from the user profile data.
8. The method of claim 1, further comprising: receiving a selection of the media item; and generating a message that includes a display of the media item.
9. The method of claim 1, wherein the historical interaction data includes at least one of: a length of time the first user account and second user account have been connected on a social networking platform; a frequency of bidirectional communications; or a recency of communications between the first user account and the second user account.
10. A system comprising: a memory; and at least one hardware processor coupled to the memory and comprising instructions that causes the system to perform operations comprising: identifying a first user account and a second user account based on user profile data; determining a characterization of a relationship between the first user account and the second user account based on historical interaction data between the first user account and the second user account; causing display of a notification at a client device associated with the first user account; receiving an input that selects the notification from the client device; responsive to the input that selects the notification, generating a media item comprising media properties selected based on the characterization of the relationship between the first user account and the second user account; and presenting the media item at the client device.
11. The system of claim 10, wherein the historical interaction data comprises bidirectional communications between the first user account and the second user account.
12. The system of claim 10, wherein the notification includes a graphical avatar that identifies the second user account.
13. The system of claim 10, wherein the media item comprises a first graphical avatar representing the first user account and a second graphical avatar representing the second user account.
14. The system of claim 10, wherein presenting the media item includes: presenting a media collection that comprises a plurality of media items generated based on the first user account and the second user account, the plurality of media items including the media item.
15. The system of claim 10, wherein determining the characterization of the relationship includes: analyzing timestamps associated with the historical interaction data; and determining a frequency of interactions between the first user account and the second user account based on the timestamps.
16. The system of claim 10, wherein the media properties are selected based on shared interests between the first user account and the second user account identified from the user profile data.
17. The system of claim 10, further comprising: receiving a selection of the media item; and generating a message that includes a display of the media item.
18. The system of claim 10, wherein the historical interaction data includes at least one of: a length of time the first user account and second user account have been connected on a social networking platform; a frequency of bidirectional communications; or a recency of communications between the first user account and the second user account.
19. A non-transitory machine-readable storage medium comprising instructions that, when executed by one or more processors of a machine, cause the machine to perform operations comprising: identifying a first user account and a second user account based on user profile data; determining a characterization of a relationship between the first user account and the second user account based on historical interaction data between the first user account and the second user account; causing display of a notification at a client device associated with the first user account; receiving an input that selects the notification from the client device; responsive to the input that selects the notification, generating a media item comprising media properties selected based on the characterization of the relationship between the first user account and the second user account; and presenting the media item at the client device.
20. The non-transitory machine-readable storage medium of claim 19, wherein the historical

interaction data comprises bidirectional communications between the first user account and the second user account.
